

JAYESH MOTWANI

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EDUCATION

Vellore Institute of Technology (VIT)	2024 - 2028
B.Tech in Electronics and Communication Engineering (ECE)	CGPA: 9.55

SKILLS

Programming Languages and Frameworks	Python, C, C++, ROS2, R, FastAPI, PyTorch, ScikitLearn, SciPy, MLFlow, Docker
Core Concepts	OOP, Version Control (Git), MLOps, SQL, Linux, Algorithm Design
Machine Learning	Data Science, Machine Learning, Deep Learning, Reinforcement Learning, Model Development, Data Analysis, RAG, LLMs, Optimization

PROJECTS

TRACE - AI-Based Network Anomaly Detection System [GitHub](#)

- Built a **VAE-based zero-day anomaly detection system (92% accuracy)** for network flow data.
- Integrated a **Mixture-of-Experts classifier** for attack-type identification and developed **FastAPI REST APIs** for real-time and batch inference.
- Added **feature-level explainability** and **LLM-based alert summarization** to assist security analysts in rapid forensic triage.

CheckMate Machine - Fair Play through Computer Vision and AI [GitHub](#)

- Built an end-to-end chess engine combining computer vision and classical AI.
- Trained a **CNN (97.2% accuracy)** to detect pieces from board images and reconstruct positions as **FEN**.
- Implemented a custom chess engine using **Minimax with α - β pruning** to evaluate positions and compute optimal moves, with scope for future **reinforcement learning based improvements**.

NyayaShield - Privacy-First Legal RAG Pipeline for Women & Children [GitHub](#)

- Built an end-to-end **legal aid system** using NLP and **retrieval-augmented generation (RAG)** to help women and children understand rights under Indian law.
- Fine-tuned a **BERT-based NER model (112 entity types)** for PII masking, ensuring fully local, privacy-preserving inference.
- Developed a **hierarchical legal parser** and conflict detection system (25+ provision overlaps), and deployed an **offline RAG pipeline** using HuggingFace embeddings, ChromaDB, and Ollama.

RESEARCH EXPERIENCE

CV-Weighted Gradient Curriculum (CVGC) for Neural Network Optimization (In Progress) 2025

- Proposed a novel Coeff. of Variation-based feature-level gradient conditioning method that acts as a time-varying data-driven preconditioner, improving early optimization while preserving $O(1/\sqrt{T})$ **convergence**. Demonstrated consistent gains across multiple tabular datasets and architectures (up to $\sim 85\%$ **faster convergence**), with theoretical proofs linking feature dispersion, gradient amplification, and optimizer dynamics.

An Intelligent Framework for Early Chronic Kidney Disease Prognosis (IGI Global - Scopus-indexed book chapter) 2025

- Authored a **Scopus-indexed book chapter** proposing a hybrid statistical feature-selection and ensemble-learning framework for early CKD prognosis, achieving high predictive performance using reduced clinical features.

LEADERSHIP & ACTIVITIES

- Maintainer for IEEE-VIT GitHub repo **scrape-krunch** for Hacktoberfest
- Won Model Arena deepfake-detection competition by developing a detection model, achieving the top accuracy and confidence score.